

DATABASE PROJECT REPORT

UNIVERSITY RESEARCH & LAB ASSET TRACKER

A. PROJECT IDENTITY

Item	Details
Project ID & Title	Project #1: University Research & Lab Asset Tracker
Team Name	G1
Team Members	1. Phạm An (n22dcat002@student.ptithcm.edu.vn) 2. Lâm Minh Chiến (n24dece057@student.ptithcm.edu.vn) 3. Đỗ Vương Bảo Anh (n24dece054@student.ptithcm.edu.vn)
Project Focus	Research grants, laboratory equipment, chemical inventory, personnel access control, EER specialization, and certification-based SQL Trigger integrity.

B. DOMAIN GLOSSARY & TAXONOMY

Domain Term	Database Semantic Definition	Representative Instance
Personnel	Common supertype representing people who may access university laboratories; specialized into Faculty or Student.	PER-00001
Research Lab	A university laboratory that stores equipment and chemicals and has an assigned hazard level.	LAB-001
Equipment	A physical research asset assigned to exactly one laboratory.	EQP-001
Chemical	A stored laboratory substance with quantity, location, and hazard classification.	CHM-001
Certification	A safety qualification defining an authorized hazard-access level.	CERT-High
Research Grant	A funded research project led by one Faculty PI and involving participating personnel.	GRANT-001
Access Log	An entry/exit audit record linking personnel to a lab and optionally an equipment item or chemical.	LOG-00001

1. SYSTEM SCOPE & PROBLEM STATEMENT

1.1 Problem Statement & Objectives

Universities operate multiple research laboratories serving faculty and students, using high-value equipment and chemicals while receiving funding from multiple research grants. When asset, chemical, grant, and access information is maintained in paper records or separate spreadsheets, traceability and safety control become difficult.

- It is difficult to control who is accessing hazardous chemicals or restricted laboratory areas.
- Equipment and chemicals may not be clearly associated with a specific laboratory or research activity, making audits difficult.
- There is no mandatory database-level rule ensuring that only personnel with appropriate and valid safety certifications can access hazardous materials.
- Personnel, grants, laboratories, and laboratory assets are insufficiently linked for centralized reporting.

The objective is to design a relational database named "University Research & Lab Asset Tracker" for centralized management of research grants, laboratory equipment, chemical inventory, and access logs. The central integrity challenge is to ensure that only personnel with valid, unexpired certifications may access materials or areas with the corresponding hazard level; this rule is intended to be enforced at the database layer through an SQL Trigger.

1.2 System Scope

Scope Area	Description
In Scope	Departments; Personnel (Faculty/Students); safety certifications; laboratories; equipment; chemicals; research grants; grant membership; access logs.
Out of Scope	Full university financial accounting; academic course registration; complete university HR management.

1.3 Functional Requirements

ID	Functional Requirement
FR1	Administrator registers new laboratory equipment and assigns it to a specific Lab.
FR2	Faculty members create and manage research grant records, including funding organization, grant amount, and project duration.
FR3	The system records each personnel laboratory access event with timestamp and Entry/Exit access type.
FR4	The database validates safety certification before access to hazardous laboratories or chemicals; enforcement is through an SQL Trigger.
FR5	The system tracks chemical inventory, stock quantity, storage location, and hazard classification.
FR6	Multiple Personnel members can participate in a Grant with roles such as PI, Co-PI, or Research Assistant.
FR7	The system supports audit/report queries such as unauthorized access attempts, equipment by lab, and certification-related information.

1.4 Non-Functional Requirements

ID	Requirement
NFR1 – Security	Certification and personnel identification data require strict access control; encryption at rest is recommended for sensitive fields.
NFR2 – Performance	Access-log audit queries should return in under 200 ms with up to 100,000 records.
NFR3 – Availability	The system should be available 24/7 because laboratories may be accessed outside normal working hours.
NFR4 – Data Integrity	Certification-based hazardous access control must be enforced at the database layer and not depend entirely on the application.

2. ENTERPRISE BUSINESS RULES & INTEGRITY CONSTRAINTS

2.1 Business Rules Subsystem Mapping & Traceability Matrix

Subsystem / Functional Area	Rule Range	Count	Associated Entities	Core Constraint
Personnel & Certification	BR1–BR5	5	PERSONNEL, FACULTY, STUDENT, DEPARTMENT, CERTIFICATION, PERSON CERTIFICATION	Disjoint/Partial specialization; certification validity
Laboratories, Equipment & Chemicals	BR6–BR8	3	LAB, EQUIPMENT, CHEMICAL, PERSONNEL, CERTIFICATION	1:N ownership/storage; hazard-level access rule
Research Grants	BR9–BR12	4	RESEARCH GRANT,	Exactly one Faculty

			FACULTY, PERSONNEL, GRANT_MEMBER	PI; N:N membership; amount/date checks
Access Log	BR13–BR15	3	ACCESS_LOG, PERSONNEL, LAB, EQUIPMENT, CHEMICAL, CERTIFICATION	FK integrity; Entry/Exit status; trigger-based authorization

2.2 Detailed Enterprise Business Rules Specification (BR1 – BR15)

Rule ID	Functional Category	Enterprise Database Business Rule & Integrity Constraint
BR1	Personnel Identity & Specialization	Each PERSONNEL has a unique person_id and may specialize as FACULTY or STUDENT. The specialization is disjoint and partial.
BR2	Faculty Department	Each FACULTY belongs to exactly one DEPARTMENT; a DEPARTMENT may have many FACULTY members.
BR3	Student Advisor	Each STUDENT may have at most one FACULTY advisor; a FACULTY member may advise many STUDENTS.
BR4	Personnel Certification	PERSONNEL and CERTIFICATION have an N:N relationship through PERSON_CERTIFICATION; each issuance records issue_date and expiry_date.
BR5	Certification Validity	A certification is valid only when the current date falls within [issue_date, expiry_date].
BR6	Lab Equipment	Each EQUIPMENT item belongs to exactly one LAB; a LAB may contain zero or more EQUIPMENT items.
BR7	Lab Chemical	Each CHEMICAL is stored in exactly one LAB; a LAB may store zero or more CHEMICALS.
BR8	Hazard Access	If a LAB/CHEMICAL hazard level is not 'None', only PERSONNEL with a valid certification at the corresponding or higher level may access it; enforced by a BEFORE INSERT trigger on ACCESS_LOG.
BR9	Grant PI	Each RESEARCH_GRANT has exactly one Principal Investigator, and the PI must be FACULTY.
BR10	Faculty as PI	A FACULTY member may be PI of multiple RESEARCH_GRANTS.
BR11	Grant Membership	A GRANT may have many participating PERSONNEL with roles through GRANT_MEMBER; a PERSONNEL member may participate in multiple GRANTS.
BR12	Grant Validity	Grant amount must be greater than 0 and end_date must be later than start_date.
BR13	Access Log References	Each ACCESS_LOG references an existing PERSONNEL and LAB; it may optionally reference a specific EQUIPMENT item and/or CHEMICAL.
BR14	Access Authorization	Each access event has access_type Entry or Exit; authorized_flag is determined

		from valid certification relative to the related Lab/Chemical hazard level.
BR15	Central DB Challenge	The system must not insert an ACCESS_LOG with authorized_flag = TRUE when the PERSONNEL lacks the corresponding valid certification; this will be implemented by SQL Trigger in Phase 3.

3. DATABASE DESIGN

3.1 Conceptual Model: Enhanced Entity-Relationship (EER) Diagram

The conceptual model uses EER specialization/generalization for PERSONNEL → FACULTY / STUDENT, associative entities for certification and grant membership, and relationships connecting laboratories, assets, grants, and access events. The original Phase 1 EER diagram is retained below.

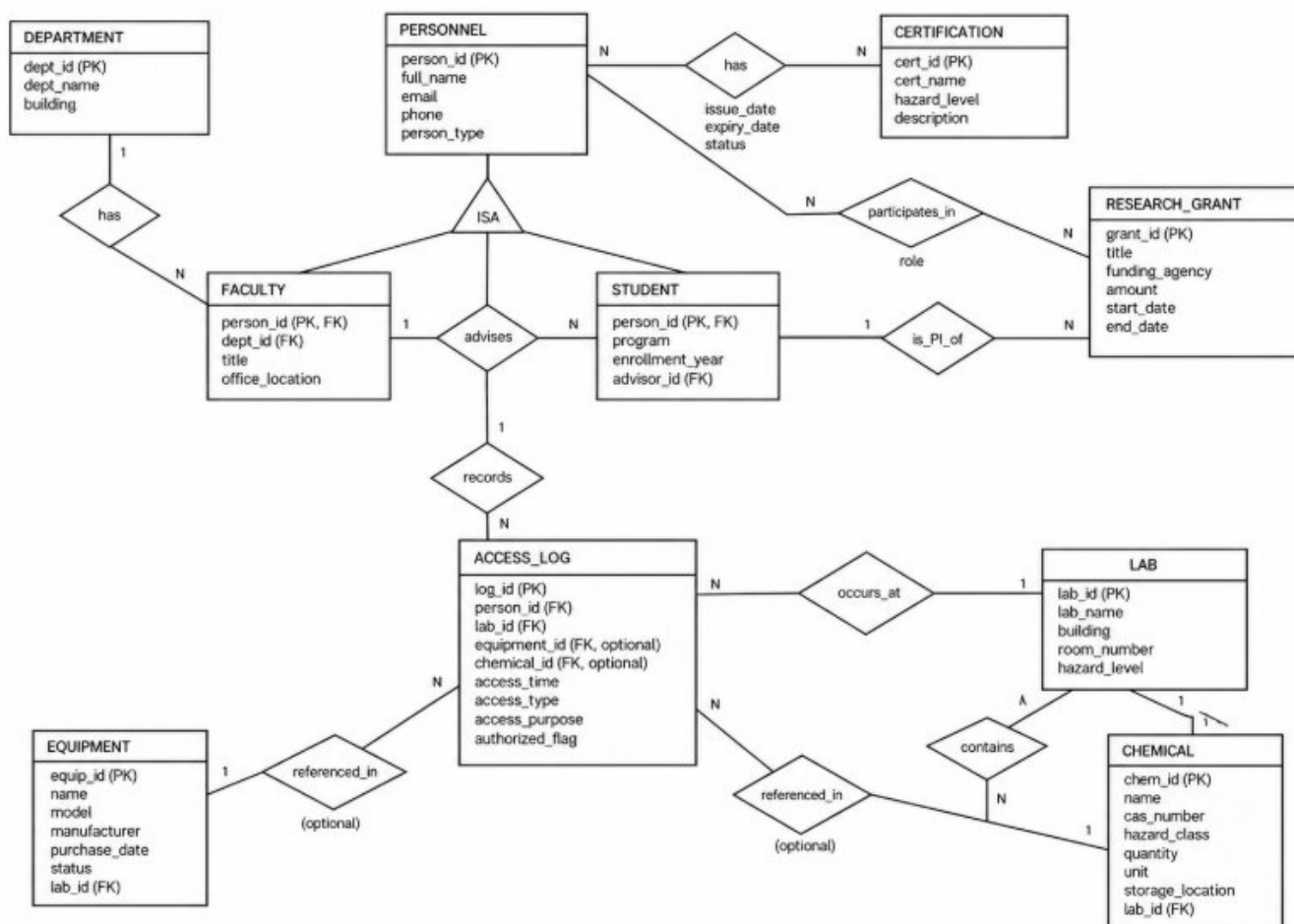


Figure 1: Comprehensive EER Conceptual Schema

3.2 Entity Overview

Entity	Primary Key (PK)	Purpose
DEPARTMENT	dept_id	Department/unit responsible for managing faculty members.
PERSONNEL	person_id	Common supertype for all personnel.
FACULTY	person_id (PK, FK)	Faculty subtype; may serve as grant PI

		and advise students.
STUDENT	person_id (PK, FK)	Student subtype; may have a Faculty advisor.
CERTIFICATION	cert_id	Safety certification type and authorized hazard-access level.
PERSON_CERTIFICATION	person_id, cert_id	Associative entity recording certification issuance and validity.
LAB	lab_id	Laboratory with its own hazard level.
EQUIPMENT	equipment_id	Equipment assigned to a specific Lab.
CHEMICAL	chemical_id	Chemical stored in a Lab with hazard classification.
RESEARCH GRANT	grant_id	Research grant with one Faculty PI.
GRANT_MEMBER	grant_id, person_id	Associative entity between Grant and participating Personnel, including role.
ACCESS_LOG	log_id	Lab entry/exit audit record with optional Equipment/Chemical reference and authorization flag.

3.3 Relationship Semantics & Referential Constraints

#	Entity Relationship	Cardinality & Participation	Business Semantics	BR
R1	DEPARTMENT → FACULTY	1 : 0..N	Each Faculty belongs to exactly one Department; a Department may manage many Faculty.	BR2
R2	FACULTY → STUDENT (advises)	1 : 0..N / Student 0..1	A Student may have at most one Faculty advisor; a Faculty may advise many Students.	BR3
R3	PERSONNEL ↔ CERTIFICATION	N : N via PERSON_CERTIFICATION	Certification issuance stores issue_date and expiry_date.	BR4–BR5
R4	LAB → EQUIPMENT	1 : 0..N	Each Equipment item belongs to exactly one Lab.	BR6
R5	LAB → CHEMICAL	1 : 0..N	Each Chemical is stored in exactly one Lab.	BR7
R6	FACULTY → RESEARCH_GRANT (PI)	1 : 0..N; Grant exactly 1 PI	Each Grant has exactly one Faculty PI; Faculty may lead multiple Grants.	BR9–BR10
R7	RESEARCH_GRANT ↔ PERSONNEL	N : N via GRANT_MEMBER	Personnel may participate in multiple Grants with assigned roles.	BR11
R8	PERSONNEL → ACCESS_LOG	1 : 0..N	Every access log references one valid Personnel member.	BR13
R9	LAB → ACCESS_LOG	1 : 0..N	Every access log references one Lab.	BR13
R10	EQUIPMENT → ACCESS_LOG	1 : 0..N (optional from log)	An access event may optionally reference one Equipment item.	BR13
R11	CHEMICAL → ACCESS_LOG	1 : 0..N (optional from log)	An access event may optionally reference one Chemical.	BR13

R12	PERSONNEL → FACULTY / STUDENT	Disjoint, Partial	A Personnel member may be Faculty or Student, but not both; either subtype may be absent.	BR1
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3.4 Design Assumptions

- A Personnel member cannot be both Faculty and Student at the same time (disjoint specialization).
- A Grant has only one PI at a time; PI change history is outside Phase 1.
- Each Access Log is associated with at most one Equipment item OR one Chemical; neither is mandatory.
- Lab, Chemical, and Certification use a consistent hazard scale (for example: None, Low, Medium, High) so the Phase 3 trigger can compare levels directly.